

DATA ANALYSIS

CUSTOMER INSIGHTS

Customer Shopping Behavior Analysis

Analyzing 3,900 transactions across product categories to uncover spending patterns, customer segments, product preferences, and subscription behavior — guiding strategic business decisions.



The Dataset

3,900 Purchases, 18 Dimensions

Customer Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

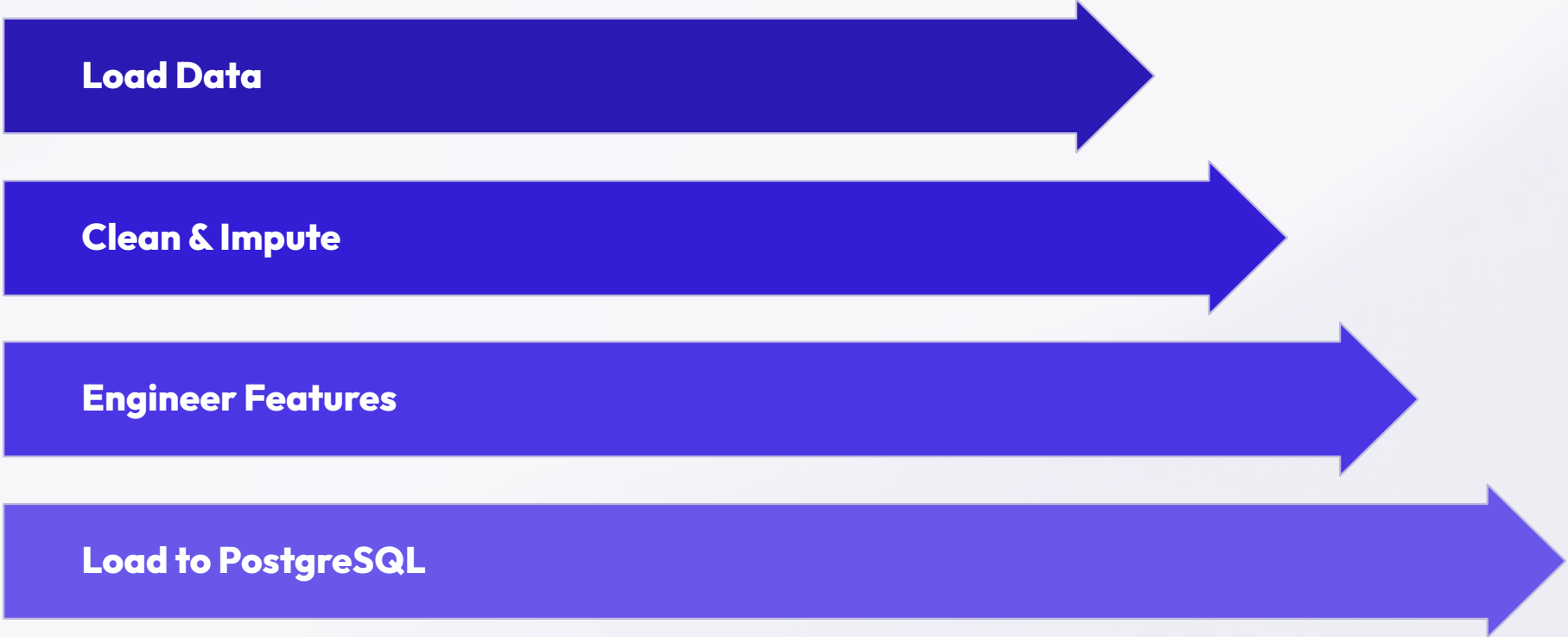
Shopping Behavior

Discount, Promo Code, Previous Purchases, Frequency, Review Rating, Shipping Type

- ❏ 37 missing values in the Review Rating column — imputed using median rating per product category.

Python & PostgreSQL Pipeline

Data Preparation & Cleaning



Load Data

Clean & Impute

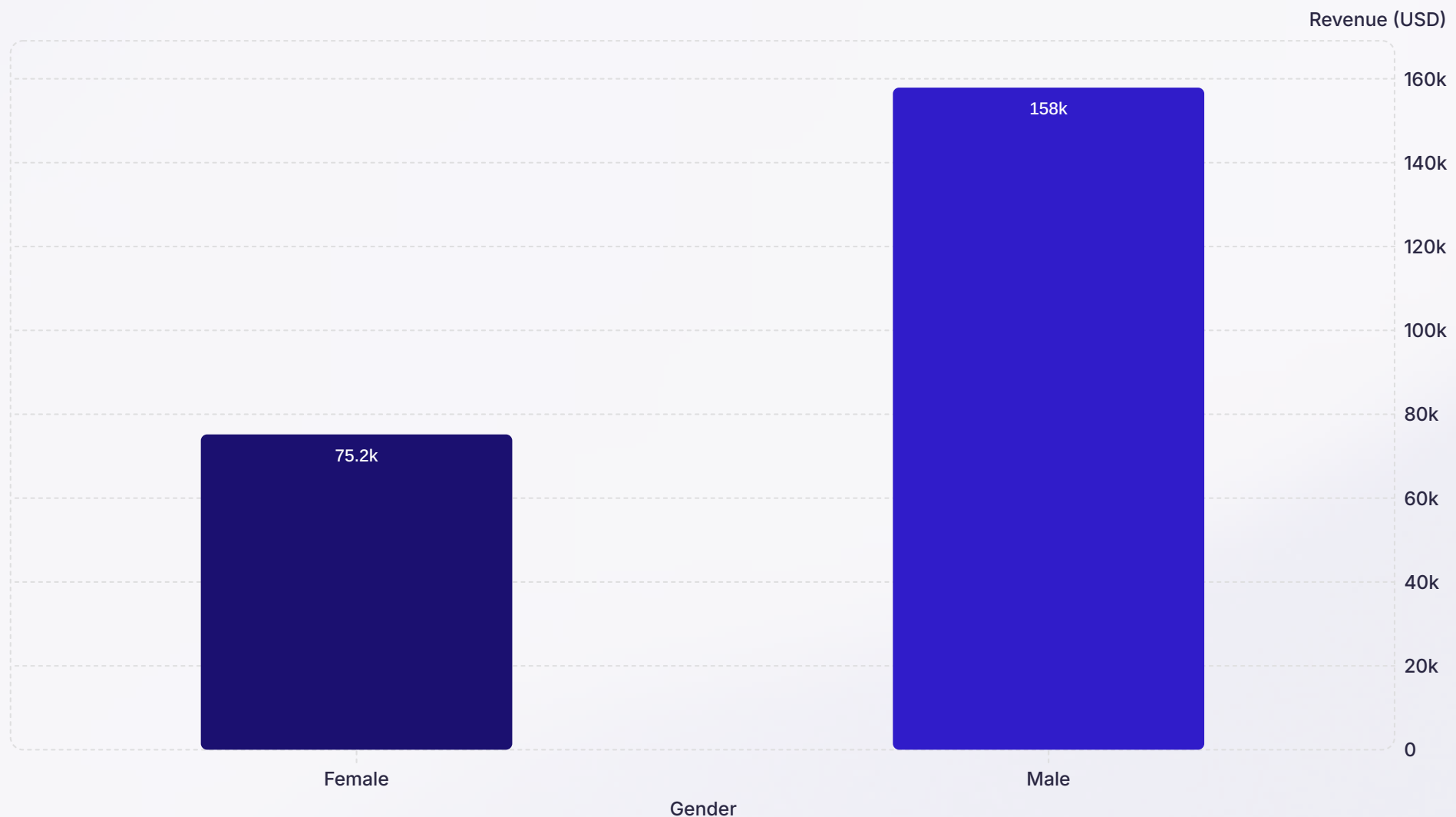
Engineer Features

Load to PostgreSQL

Columns standardized to snake case. Redundant `promo_code_used` dropped after confirming it mirrored `discount_applied`. Cleaned DataFrame loaded into PostgreSQL for structured analysis.

Key SQL Findings

Revenue & Customer Segmentation



Male customers generated **\$157,890** — more than double the **\$75,191** from female shoppers.

3,116

Loyal Customers

701

Returning

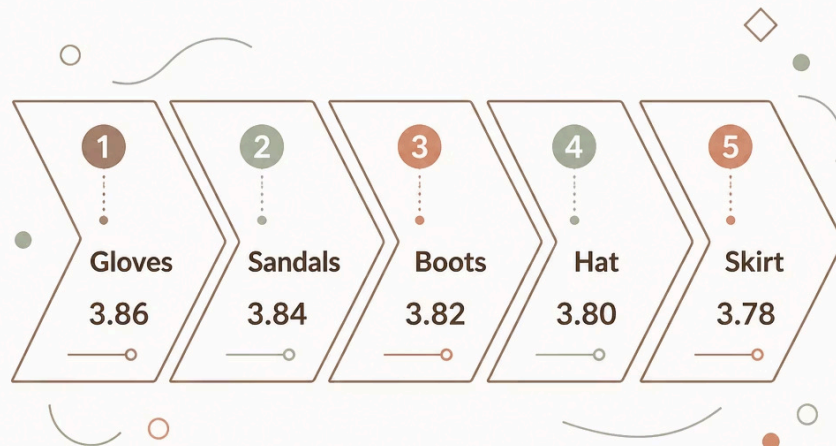
83

New

839 high-spending customers used discounts yet still spent above the average purchase amount of \$59.76.

Product & Shipping Insights

What Customers Buy & How It Ships

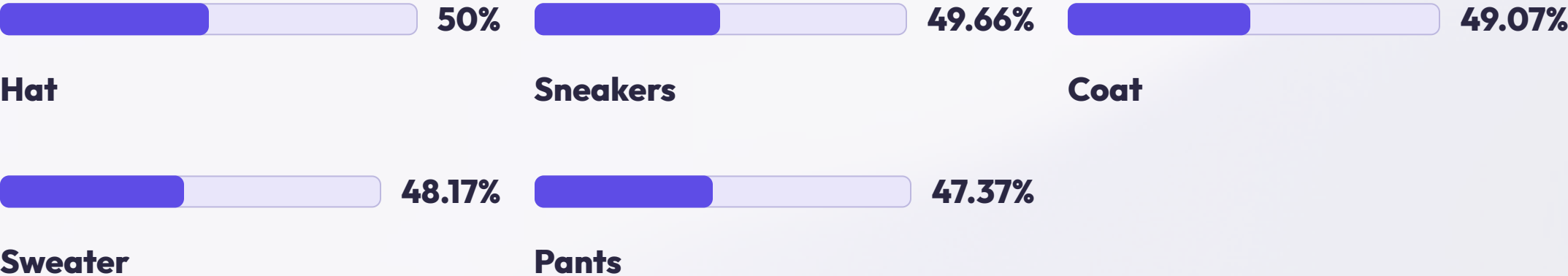


Express shipping averages **\$60.48** per purchase versus **\$58.46** for Standard — a modest but consistent premium.

Top products per category: **Jewelry** (Accessories), **Blouse** (Clothing), **Sandals** (Footwear), **Jacket** (Outerwear).

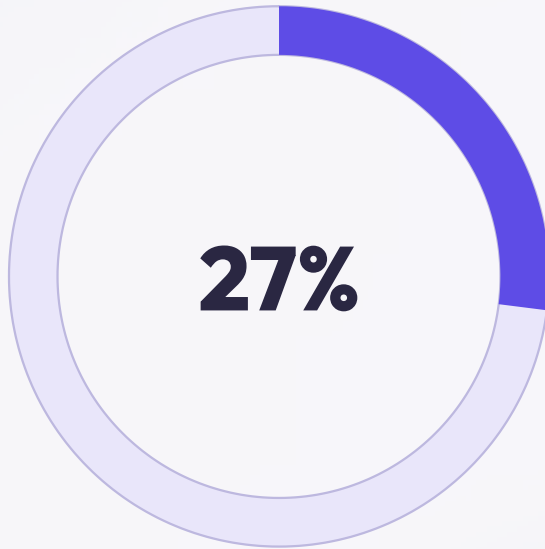
Discount-Dependent Products

Discounts Drive These Categories

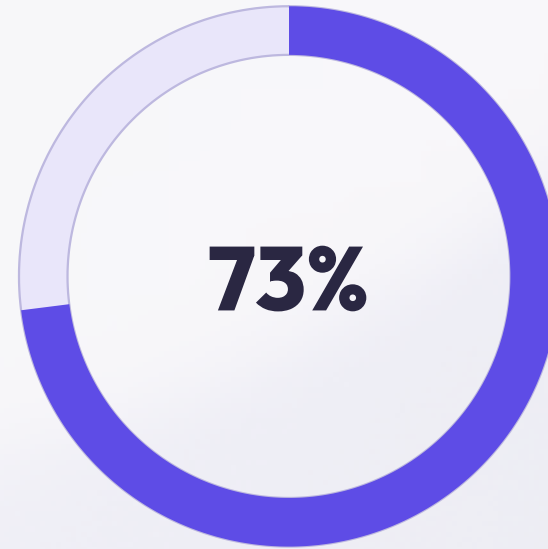


These five products see the highest share of discounted purchases — nearly half of all transactions involve a discount.

Visualizing the Customer Journey



Subscribers



Non-Subscribers

Interactive filters by gender, category, and shipping type reveal that **Young Adults** drive the most revenue (\$62,143), and **Clothing** dominates across both revenue and sales volume.

Only **958** of 3,476 repeat buyers subscribe — a clear opportunity for conversion.

Strategic Next Steps

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert non-subscribers.



Loyalty Programs

Reward repeat buyers to move them into the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control on high-discount products.



Product Positioning

Highlight top-rated and best-selling products in campaigns.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users.